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GIRIK System: Multi-Certificate Concurrent Workflow

This document explains the exact behavior of the system when **a single Job Request contains multiple, independent Certificate Types**. It outlines how different certificates can be in completely different states at the exact same time without blocking each other.

To illustrate, we will use a **Single Job Request (Job #100)** that has **three certificates**:

1. **Certificate A (Annual Inspection)**
 2. **Certificate B (Intermediate Inspection)**
 3. **Certificate C (Bottom Inspection)**
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The Core Rule of Independence

The parent **JobRequest** is just an umbrella folder. **The true state machine runs individually on each JobCertificate**. This means Certificate A can be fully completed (**ISSUED**) while Certificate C hasn't even received its documents yet (**PENDING**).

Step 1: Job Creation (The Starting Point)

Actor: Client

- **What Happens:** The client creates Job #100 and selects Annual, Intermediate, and Bottom certificates.
 - **Database State:**
 - **JobRequest #100** -> **CREATED**
 - **Certificate A (Annual):** **status** = **PENDING**
 - **Certificate B (Intermediate):** **status** = **PENDING**
 - **Certificate C (Bottom):** **status** = **PENDING**
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Step 2: Asynchronous Document Verification

Actor: Technical Officer (TO) / GM

- **What Happens:** The client only uploads documents for Certificate A. The TO reviews and verifies ONLY Certificate A's documents.
 - **Database State:**
 - **Certificate A:** **DOCUMENT_VERIFIED** (*Ready for next step*)
 - **Certificate B:** **PENDING** (*Blocked, waiting for docs*)
 - **Certificate C:** **PENDING** (*Blocked, waiting for docs*)
 - **JobRequest #100** -> Automatically syncs to **IN_PROGRESS** because at least one certificate started moving.
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Step 3: Independent Surveyor Assignment

Actor: General Manager (GM)

- **What Happens:** The GM wants to assign a surveyor. They use the **Assign Surveyor to Certificate** feature (not bulk) to assign Surveyor John to Certificate A.
- **Backend Guard in Action:** If the GM tried to use "Bulk Assign" for the whole job, the backend would throw a **400 Error** because Certificates B & C are still **PENDING**.
- **Database State:**

- **Certificate A:** Remains **DOCUMENT_VERIFIED**, but now **assigned_surveyor_id** = John.
 - *Eager Action:* A **Survey** record is instantly created for Certificate A (**NOT_STARTED**).
 - **Certificate B & C:** Remain **PENDING**. No survey records exist for them yet.
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Step 4: Staggered Survey Authorization

Actor: Technical Manager (TM)

- **What Happens:** TM reviews Certificate A and clicks "Authorize Survey". Meanwhile, the client finally uploads documents for Certificate B, and the TO verifies them.
 - **Database State:**
 - **Certificate A:** **SURVEY_AUTHORIZED** (*John can now start his survey*).
 - **Certificate B:** **DOCUMENT_VERIFIED** (*Ready for surveyor assignment*).
 - **Certificate C:** **PENDING**.
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Step 5: Parallel Survey Execution

Actor: Surveyor(s)

- **What Happens:** Surveyor John opens his mobile app. He sees ONLY the survey for Certificate A, because it is the only one authorized. He clicks "Start Survey", uploads photos, and submits.
 - **Database State:**
 - **Certificate A:** **SURVEY_AUTHORIZED** (Under the hood, its **Survey** record moves through **STARTED** → **PROOF_UPLOADED** → **SUBMITTED**).
 - **Certificate B:** Still **DOCUMENT_VERIFIED** (Waiting for GM to assign a surveyor).
 - **Certificate C:** Still **PENDING**.
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Step 6: Mixed Final Reviews & Reworks

Actor: Technical Manager (TM) / GM

- **What Happens:** TM reviews Surveyor John's report for Certificate A. It's perfect, so TM approves it. Meanwhile, GM assigns Surveyor Mike to Certificate B, and TM authorizes it.
 - **Database State:**
 - **Certificate A:** **SURVEY_DONE** (Survey is **FINALIZED**).
 - **Certificate B:** **SURVEY_AUTHORIZED** (Survey is **NOT_STARTED**).
 - **Certificate C:** **PENDING**.
 - **What if Mike messes up Certificate B?**
 - TM clicks "Request Rework" on Certificate B.
 - **Certificate B** goes backwards to **REWORK_REQUESTED**.
 - *Crucial Point:* This does **NOT** affect Certificate A, which remains safely **SURVEY_DONE**.
 - GM can now use the "Reassign" button specifically on Certificate B to fire Mike and hire Surveyor Sarah, without touching Certificate A.
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Step 7: Staggered Certificate Generation

Actor: Admin

- **What Happens:** Since Certificate A is **SURVEY_DONE**, the Admin clicks "Generate PDF Certificate" for Certificate A. They hand the PDF to the client.
- **Database State:**
 - **Certificate A:** **ISSUED** (Completely finished terminal state).
 - **Certificate B:** Still undergoing rework (**REWORK_REQUESTED**).
 - **Certificate C:** Still waiting for documents (**PENDING**).
 - **JobRequest #100:** Remains **IN_PROGRESS**.
- **The Final Curtain:** Only when Certificate B and Certificate C eventually reach **ISSUED** (or **REJECTED**), will the **lifecycle.service** auto-sync the parent

JobRequest #100 to its final, global CERTIFIED status.